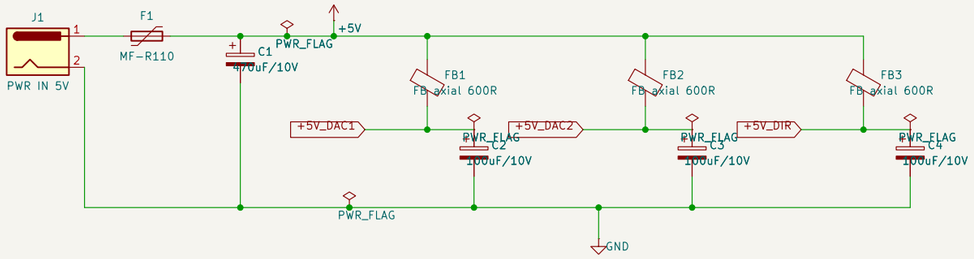
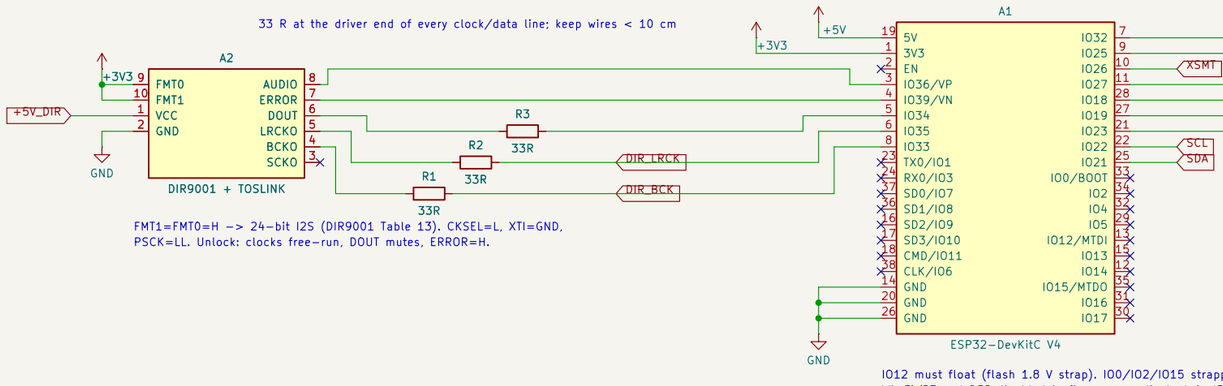
Power in (5 V class II adapter -> PTC -> 470 uF bulk; ferrite + 100 uF per module rail. Star ground at the jack.)



FB1-3: axial bead on lead, ~600 R @ 100 MHz, DCR < 0.3 R. Each module keeps its own 100 nF + 10 uF at the pin. Never power from a PC USB port.

Signal chain (TOSLINK -> DIR9001 clock recovery -> ESP32 I2S slave on both ports -> PCM5102A x2 in 3-wire BCK-PLL mode -> line out)

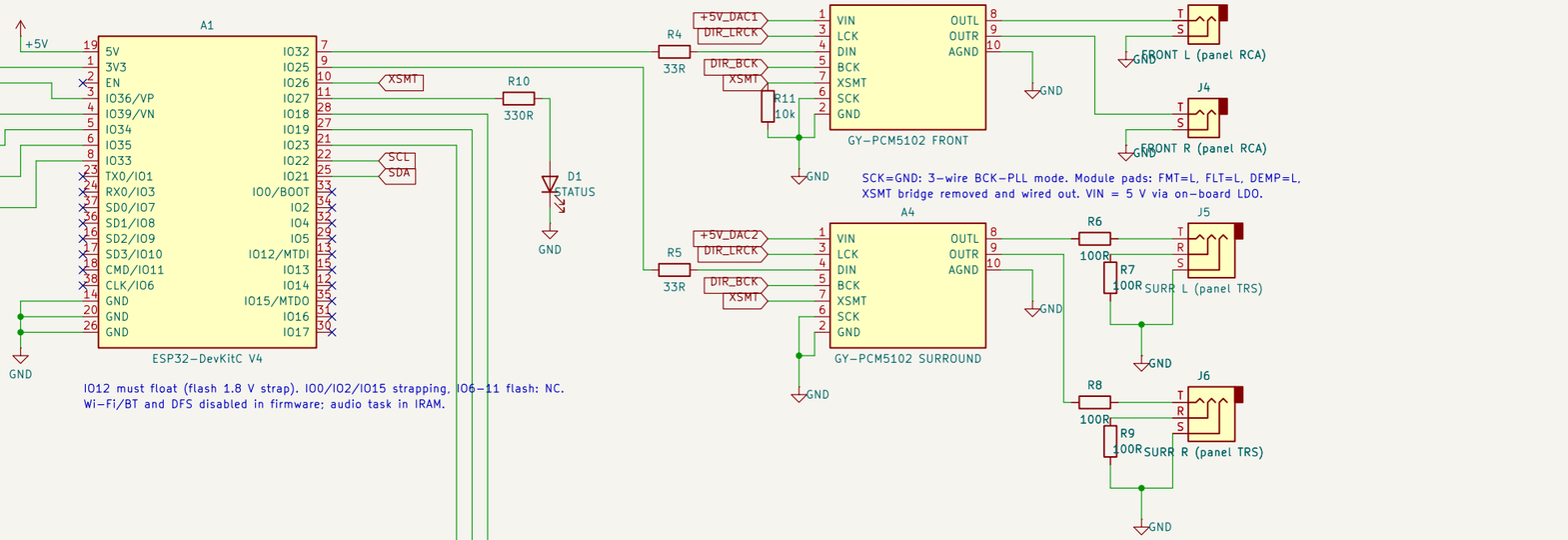
33 R at the driver end of every clock/data line; keep wires < 10 cm



FMT1=FMT0=H -> 24-bit I2S (DIR9001 Table 13). CKSEL=L, XTI=GND, PSCK=LL. Unlock: clocks free-run, DOUT mutes, ERROR=H.

IO12 must float (flash 1.8 V strap). IO0/IO2/IO15 strapping, IO6-11 flash: NC. Wi-Fi/BT and DFS disabled in firmware; audio task in IRAM.

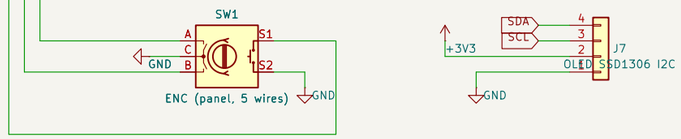
Fronts: panel RCA jacks on screw terminals, short cable to the Fosi amp. Module has 470 R + 2.2 nF out filter.



SCK=GND: 3-wire BCK-PLL mode. Module pads: FMT=L, FLT=L, DEMP=L, XSMT bridge removed and wired out. VIN = 5 V via on-board LDO.

Panel TRS jacks on screw terminals, impedance-balanced to the Eris 3.5: 100 R tip, 100 R ring to GND.

User interface (rotary encoder on PCNT with glitch filter, internal pull-ups; SSD1306 OLED on I2C with pull-ups on the module)



Sheet: /		
File: surround.kicad_sch		
Title: ESP32 4.0 surround processor — carrier board		
Size: A3	Date:	**Rev:**
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